

Australian model of a space station module for astronaut training

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Abstract

The International Space Station (ISS) took ten years to assemble in Low Earth Orbit, with NASA (United States) and Roscosmos (Russia) contributing equally in the beginning stages of construction. From 2010, the ISS entered the operations phase and included components and habitable modules from JAXA (Japan), ESA (Europe) and CSA (Canada). The station is used as an orbital laboratory and has surpassed its operational lifetime with initial plans to de-orbit in 2024 now extended to 2028.

On Earth, when astronauts are allocated to an ISS Expedition crew, they receive mission specific training at a number of facilities provided by the 5 main space agencies. These training facilities are full-scale mock-ups of the habitable modules of the ISS. The ESA Columbus module is 6 metres long and 4.5 metres in diameter, therefore the training model must be housed in a large facility. The training hall at the European Astronaut Centre houses a training facility of the Columbus and Node 2 modules, the Automated Training Vehicle (ATV) and a 10-metre dive pool.

Saber Astronautics invited Australians to apply to their astronaut program which they have established with Axiom Space. There are currently no full-scale space station training modules in Australia. We have designed a model of such a training module to guide the future construction of a full-size station module. We have constructed a 1:8 size model of the training model from laser-cut aluminium to demonstrate the concept. We have also designed a full-scale international standard payload rack for external users to test their payload operation prior to launch to the ISS and train astronauts how to install and activate the payload. We detail here the systems requirements for payloads launching to space stations such as ISS, Axiom space station or Orbital Reef.

Keywords: Astronaut training, International Space Station, Payload rack, Space station training model, engineering model.

Introduction

The International Space Station (ISS) represents the culmination of global efforts in Low Earth Orbit, serving both as a research laboratory and an advanced platform where astronaut crews live and work for months at a time [1]. Supported by NASA, Roscosmos, JAXA, ESA, and CSA, the ISS has enabled significant advancements in scientific exploration and the study of human physiological changes in reduced gravity [2]. Construction began in 2000, requiring ten years before completion, and since 2010 the ISS has been permanently crewed by humans. Its lifetime was originally set until 2024, however, this has since been extended to 2028.

In preparation for the deorbiting of the ISS, NASA have contracted independent company, Axiom Space, to build a number of habitable modules which will be launched into space and dock with the ISS. When the ISS reaches end-of-lifetime, in ~2028, the ISS will be undocked and will conduct a controlled re-entry into Earth's atmosphere, leaving the Axiom modules to remain in orbit operating independently [3].

In 2022, Saber Astronautics announced the development of the “first formal program to enable Australians and Australian industry to, via Axiom, access the Station’s orbiting microgravity laboratory and develop a new generation of space-developed products for human health, materials, electronics, cleantech, and more” [4]. The call was designed to invite commercial companies to propose experiments and equipment that would operate on board Axiom station. In return for this opportunity, Australia would have sufficient ‘buy-in’ to justify the launch of an Australian citizen to operate this equipment.

Earth-based analogue modules of the ISS, built on a 1:1 scale, play a vital role in astronaut training and equipment testing. This is evident in facilities such as the Space Vehicle Mockup Facility (SVMF) at Johnson Space Center (JSC) (shown in Figure 1). However, the absence of similar resources in Australia has restricted its contributions to the space industry, highlighting an infrastructure gap that limits the nation’s capacity to engage in astronaut training and equipment validation effectively [5].



Figure 1. The Space Vehicle Mockup Facility (SVMF) at Johnson Space Center (JSC) used for training contractors and International Space Station (ISS) International Partners, including ISS crew. Credit: NASA.

ISS habitable modules are very large; 4.5 metres in diameter and up to 6 metres long. The modules that have been launched into space must have the capability to be fully pressurised and to maintain a steady temperature and breathable atmosphere in order for the crew to live and work without the need for spacesuits. The *training* models of these modules on Earth need none of these features. They remain located in a large warehouse/hangar-sized building and use ambient pressure and temperature. Some aspects of the modules are high fidelity; i.e. the equipment such as valves and tubing will be exact replicas of those found in the orbiting module, whilst other aspects may be very low fidelity; e.g. the walls may be built out of plastic where the outer hull of the ISS modules are manufactured from aluminium.

Inside the habitable modules of the ISS, equipment racks are positioned in such a way that they make the inside look cubic, effectively turning a cylindrical module into a rectangular room. Therefore, the racks are D-shaped with a curved back – which faces the walls – and the flat front creates a flat wall on the interior. Individual racks are 2 metres tall, 1.6 metres wide and 1 metre deep. There is a small (< 40 cm) gap behind the mid-point of the racks, at the maximum diameter of the modules, which is used to run the cables, ducts and piping that supply both the

racks and the modules themselves [4]. These equipment racks are termed International Standard Payload Racks (ISPR) and contain systems equipment, scientific experiments or are used for stowage. A schematic of the European Space Agency module, 'Columbus' [5] is provided in figure 2 to illustrate the layout.

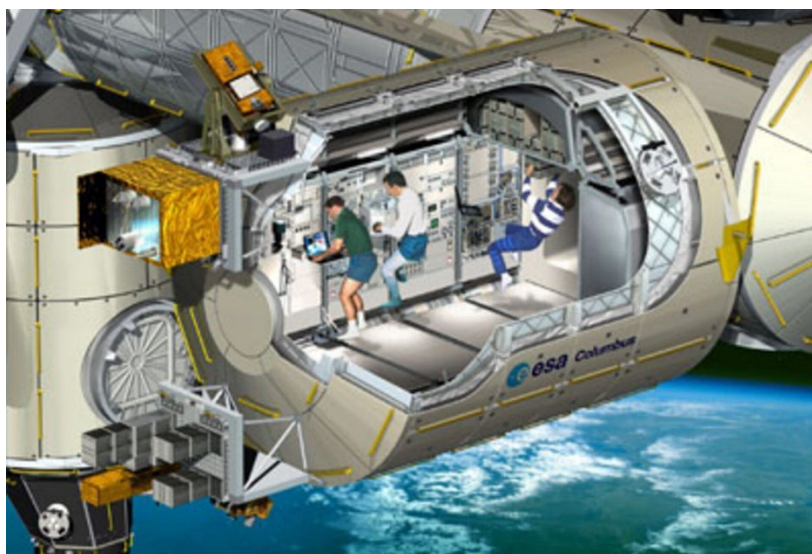


Figure 2. Europe's laboratory on the International Space Station. Credit: ESA.

One issue facing Australian experiment developers is a suitable platform in which to test their hardware before launch into space either as sounding rocket payloads or ISS payloads. The construction of a test facility in Australia would provide an interface for such integration and operations testing. Validation and verification are the most critical steps for the success of a space mission, and these steps are most often rushed or overlooked altogether. Therefore, this paper highlights the steps taken to develop a full-scale space station module with high fidelity interfaces and operational capability.

Methods

Three main methods were utilised in this project; Computer-Aided Design (CAD) for the production of technical drawings, prototype construction using a range of materials and mechanical engineering, and Finite Element Analysis (FEA) to verify material choices and structural integrity.

Computer-Aided Design

Using SolidWorks as the software of choice, technical drawings for the prototypes were produced from CAD component assemblies.

Multi-phase prototyping

The engineering approach for developing this space station analogue module involved a multiphase prototyping strategy, each phase building upon the last to refine functionality, structural stability, and modularity. 3D-printing using ABS was used for plastic prototypes and a combination of laser-cut and water-cut aluminium parts were used for a metal prototype.

Modularity and Structural Refinements

A key aspect of the project was the design and development of modular components to facilitate assembly, transport, and adaptability. Modular sections were engineered with custom-designed brackets and trusses to securely connect individual segments.

The initial modular prototype included 6,155 parts, underlining the complexity of the design. This design was simplified by eliminating modularity and incorporating planar truss components, reducing the total part count to 2,204, including fastenings. This streamlined the

manufacturing and assembly processes while retaining structural integrity and functionality, ensuring the project remained on schedule and within budget constraints. Each sheet metal panel and truss structure were manufactured via laser cutting, and long thin parts such as beams and trusses were manufactured via water jet cutting. Fastening was achieved using M3 bolts for non-permanent assembly and rivets for permanent connections.

Finite Element Analysis

To verify material choices and structural integrity, Finite Element Analysis (FEA) was conducted using ANSYS software, focusing on load-bearing components and modular stability.

Results

Habitable Module Prototypes

Multiple CAD drawings were produced, and a 3D-model of a 1:16 scale, single generic ISS habitable module was printed in plastic. This model was used to confirm relative dimensions and overall design choices relating to number of racks, hatch placement and novel design features i.e. cupola on one end. The plastic prototype created the larger features well; however, smaller features were too small for full integrity. Therefore, a larger scale model was required, with higher fidelity, using a better material than plastic. Figure 3 is the CAD drawing of a 1:8 scale model of the module manufactured from aluminium. This prototype comprised 2204 separate components.

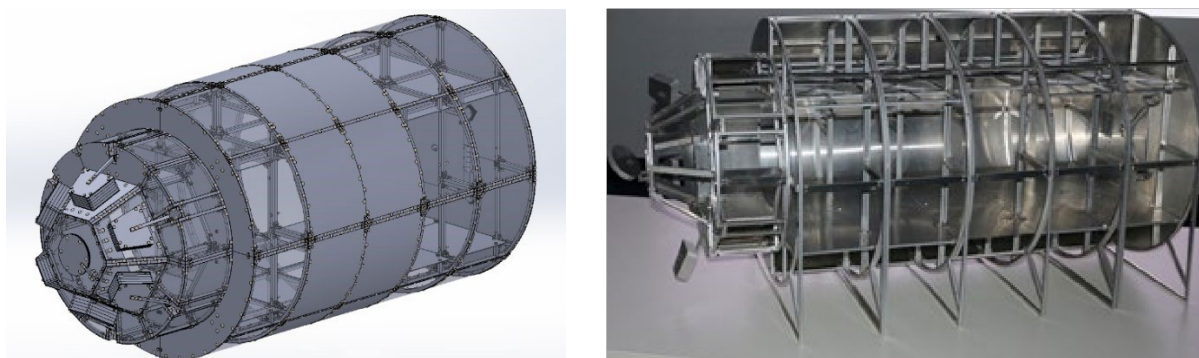


Figure 3. The RMIT University space module 1:8 scale model, with CAD drawing (left) and prototype manufactured from aluminium (right).

Aluminium alloy 6061 was chosen for the full-scale frame due to its high load capacity, low density, and cost-effectiveness when compared to 7XXX series alloys. FEA results demonstrate minimal deformation under exaggerated load stresses (figure 4). This finding confirmed aluminium's suitability as a durable, lightweight alternative to heavier materials like steel, aligning with the module's requirements for portability and ease of assembly. Functional evaluations examined the performance of components such as doors and racks, while redundancy tests simulated mechanical failures to assess resilience and repair efficiency. These results informed design refinements that ensured the final model met the rigorous standards required for astronaut training and experimental testing.

Equipment Racks

A generic design was chosen for the equipment racks, and material of magnesium alloy was selected for a balance of strength and lightweight design for the support structure (figure 5). The frame was designed so that racks could be fully modular and reconfigurable, allowing them to accommodate a wide range of research applications. Four experimental chambers were selected for integration into one rack and designed to operate either simultaneously or independently. The experimental chambers allow experiments in life sciences, physical sciences, and multi-functional modular experiments to occur at the same time, making this rack versatile and easily configurable.

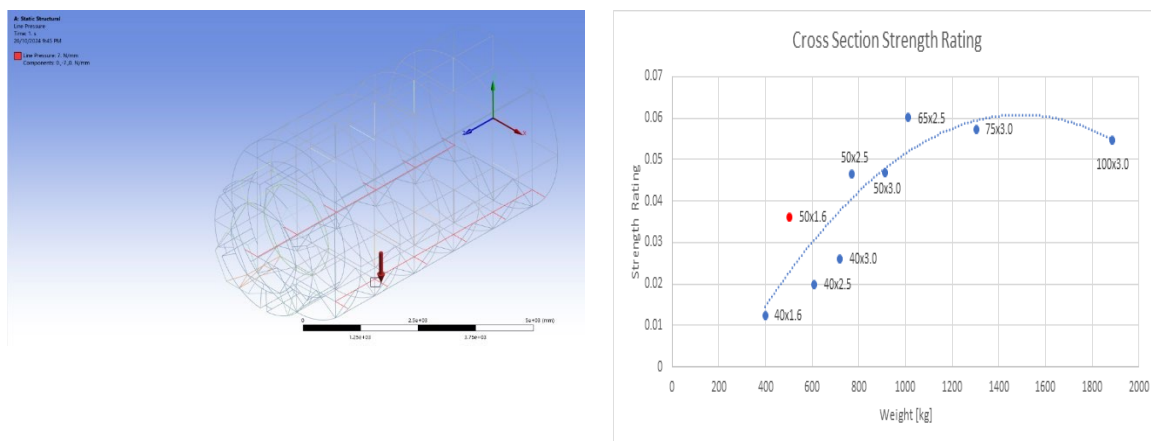


Figure 4. (a) FEA analysis

(b) Cross Section Strength Rating

Mechanical specifics of the rack and the experimental chambers it houses, along with the scientific capabilities they support, are summarised in table 1.

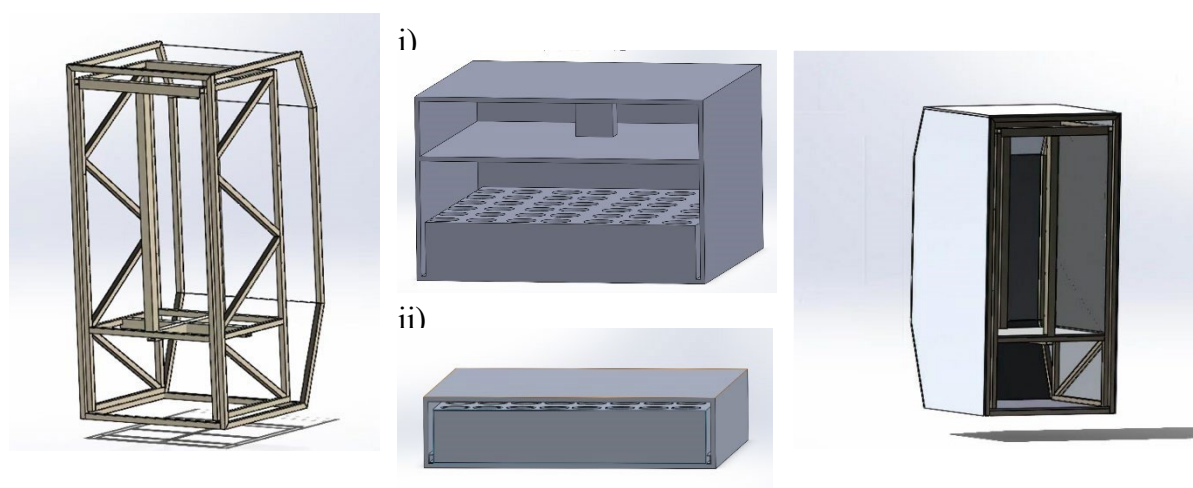


Figure 5. (a) Rack frame

(b) Fridge (Prototype(i)), (c) Rack with outer panels final design (ii)

When the equipment rack is fully loaded with all four experiment chambers, the total mass is 190.32 kg. If a full-scale habitable module were to be built on Earth, and this module were to contain multiple experiment racks, then any floor would have to be load-tested and approved to manage the total mass.

Table 1. Mechanical specifics of the RMIT Multi-purpose Equipment Rack and associated scientific capabilities.

	Mass (kg)	Material	Power (W)	Temp (C)
Rack (empty)	76.12	Mg alloy	N/A	N/A
Fridge	33	Aluminium	300-800	-40 to +70
Cold-welder	40	Aluminium	3300	Ambient
Plant bed	25.2	Aluminium	90	20 to 24
Modular drawer	16	Aluminium	15	Ambient

A full-scale wooden habitable module was designed, and its mass was 108.19 kg. Whilst this could be managed from a floor load-bearing perspective, the sheer size of the fully constructed module makes storage and or presentation complicated as a room large enough to house the module must be available.

Conclusion

The development of this Australian space station analogue module marks a pivotal step toward fulfilling the nation's need for a dedicated astronaut training and equipment testing facility. The module's ISS-inspired design, combined with its modular ISPR framework, enables it to support a wide variety of research and training applications, positioning Australia as an emerging presence in the international space community.

The prototypes produced in this project include a 1:16 scale model of a space station module made out of 3D-printed plastic, and a 1:8 scale model of a module made from laser-cut aluminium. This higher-fidelity metal model, almost 1 metre in length, can be fitted with electrical connections and used as a functional model for systems engineering testing and training purposes.

CAD drawings for an ISS International Standard Payload Rack (ISPR) have been produced, which can house up to four novel science experiments including a fridge, cold-welding station, plant-growing pad and modular drawer unit. This allows users to test experiments in life sciences, physical sciences, and material durability using the same connectors and communications protocol as encountered on the ISS.

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